STA 243: Homework 2

- Homework due in Canvas: 05/06/2025 at 9:00PM. Please follow the instructions provided in Canvas about homeworks, carefully.

1. (**5 Points**) Prove that a differentiable function $f(\theta) : \mathbb{R}^d \to \mathbb{R}$ is convex if and only if

$$f(\theta_2) \geq f(\theta_1) + \nabla f(\theta_1)^\top (\theta_2 - \theta_1)$$

Hint: Think of 1-dimensional case and extend the intuition to d-dimensional case.

Solution: ($\Rightarrow$) Assume f is convex. Fix arbitrary $\theta_1, \theta_2 \in \mathbb{R}^d$ randomly and denote $g(t) = f\big(\theta_1 + t(\theta_2 - \theta_1)\big)$, $\forall t \in [0,1]$. Convexity of f implies g is convex on $[0,1]$, then we have

$$g(t) \;\leq\; (1-t)\, g(0) \;+\; t\, g(1) \quad \Rightarrow \quad \frac{g(t) - g(0)}{t} \;\leq\; \frac{g(1) - g(t)}{1-t}\,.$$

Letting $t \to 0^+$, then on the LHS, we have

$$g'(0) \;\leq\; g(1) - g(0).$$

Since

$$g(0) = f(\theta_1), \quad g(1) = f(\theta_2), \quad g'(t) = \nabla f\big(\theta_1 + t(\theta_2 - \theta_1)\big)^\top (\theta_2 - \theta_1),$$

we obtain

$$\nabla f(\theta_1)^\top (\theta_2 - \theta_1) \;\leq\; f(\theta_2) - f(\theta_1),$$

as required.

($\Leftarrow$)

Conversely, suppose

$$f(\theta_2) \;\geq\; f(\theta_1) \;+\; \nabla f(\theta_1)^\top (\theta_2 - \theta_1) \quad \forall\, \theta_1, \theta_2 \in \mathbb{R}^d.$$

Fix $\theta_1, \theta_2 \in \mathbb{R}^d$ and $\alpha \in [0,1]$ randomly. Define $\theta_3 = \alpha\,\theta_2 + (1-\alpha)\,\theta_1$. Applying the first-order condition at θ_3 with (θ_1, θ_3) and (θ_2, θ_3) gives

$$f(\theta_1) \;\geq\; f(\theta_3) \;+\; \nabla f(\theta_3)^\top (\theta_1 - \theta_3), \tag{1}$$

$$f(\theta_2) \;\geq\; f(\theta_3) \;+\; \nabla f(\theta_3)^\top (\theta_2 - \theta_3). \tag{2}$$

Multiply (1) by $(1-\alpha)$ and the (2) by α, then add them together:

$$(1-\alpha) f(\theta_1) + \alpha f(\theta_2) \;\geq\; f(\theta_3) + \nabla f(\theta_3)^\top \big((1-\alpha)(\theta_1 - \theta_3) + \alpha(\theta_2 - \theta_3)\big).$$

Observe that

$$\begin{aligned}(1-\alpha)(\theta_1 - \theta_3) + \alpha(\theta_2 - \theta_3) &= (1-\alpha)\big(\theta_1 - \big((1-\alpha)\theta_1 + \alpha\theta_2\big)\big) \\ &\quad + \alpha\big(\theta_2 - \big((1-\alpha)\theta_1 + \alpha\theta_2\big)\big) \\ &= 0.\end{aligned}$$

After rearranging terms, we get

$$(1-\alpha) f(\theta_1) + \alpha\, f(\theta_2) \;\geq\; f(\theta_3) \;=\; f\big((1-\alpha)\theta_1 + \alpha\theta_2\big),$$

which is exactly the definition of convexity.

$\square$

2. (**20 Points**) The origin of the dataset `housingprice.csv` we will use in this question is from the Coursera open course Machine Learning Foundations: A Case Study Approach by Prof. Carlos Guestrin and Prof. Emily Fox. Load the training data `train.data.csv` and testing data `test.data.csv`. We'll build our regression model on the training data and evaluate the model on the testing data.

   (a) Build a linear model (you are free to use any Python package for this) on the training data by regressing the housing price on these variables: `bedrooms`, `bathrooms`, `sqft_living`, and `sqft_lot`. What's the R^2 of the model on training data? What's the R^2 on testing data?

Dataset	R^2
Train	0.51011
Test	0.50500

Table 1: R^2 of regression with no interaction term

   (b) The image below is Bill Gates' house. Load the file `fancyhouse.csv` to obtain the features of the house. Guess the price of his house using your linear model. Do you think the predicted price is reasonable?

Figure 1: Image from Wikipedia Commons

Bedrooms	Bathrooms	`sqft_living`	`sqft_lot`
8	25	50000	225000

Table 2: Bill Gates' house features

   The predicted price of Bill Gates' house is **$15,436,769.54**. I think the price is reasonable.

   (c) Let's continue to improve the linear model we have. Instead of throwing only the raw data into the statistical model, we might want to use our intuition and domain expertise to extract more meaningful features from the raw data. This step is called feature engineering. Using meaningful features in the model is often crucial for successful data analysis. Add another variable by multiplying the number of bedrooms by the number of bathrooms, which describes the combined benefit of having more bedrooms and bathrooms. Add this variable to the linear model we have in Part (a). What's the R^2 of the new model on the training data and testing data respectively?

Dataset	R^2
train	0.51735
test	0.51036
Predicted price	\$18,607,312.89

Table 3: R^2 of regression with interaction feature

(d) Perform parts (a), (b) and (c) above without using any in-built function in Python (i.e., any packages that are related directly to linear regression), but by using `gradient descent algorithm` on the sample-based least-squares objective function, to estimate the OLS regression parameter vector. How does your result compare to the result from previous part ? Note that you have to set the step-size parameter appropriately for this method.

	OLS	OLS, GD	Interaction	Interaction, GD
X1	-0.15147017	-0.15147017	-0.32877205	-0.32877205
X2	0.00773629	0.00773629	-0.23326364	-0.23326364
X3	0.79183280	0.79183280	0.77192759	0.77192759
X4	-0.04514359	-0.04514359	-0.04491517	-0.04491517
X1X2			0.38964848	0.38964848

Table 4: Estimated coefficients under various models

Model	Train	Test
OLS	0.51011	0.50499
OLS, GD	0.51011	0.50499
Interaction	0.51735	0.51054
Interaction, GD	0.51735	0.51054

Table 5: R^2 for GD

We fit the ordinary least squares estimator

$$\beta^* \;=\; \arg\min_{\beta\in\mathbb{R}^4} (y - X\beta)^\top (y - X\beta)$$

by applying gradient descent to the equivalent quadratic form. Define

$$\bar{\beta} = (X^\top X)^{-1} X^\top y, \quad A = 2\,X^\top X, \quad f(\beta) = \tfrac{1}{2}(\beta - \bar{\beta})^\top A(\beta - \bar{\beta}),$$

so that

$$\nabla f(\beta) = X^\top \big(X\beta - y\big).$$

A convenient constant step size is

$$\eta = \frac{2}{\lambda_{\max}(A) + \lambda_{\min}(A)}.$$

We initialize $\theta^{(0)} \sim \mathcal{N}(0,1)$, and run the gradient descent algorithm which is given below.

Algorithm 1 Gradient Descent for OLS

Require: Initial guess $\theta^{(0)} \in \mathbb{R}^d$, step size $\eta > 0$, tolerance $\tau > 0$, max iter. T
Ensure: Approximate solution $\theta^{(t)} \approx \beta^*$

```
for t = 0, 1, ..., T − 1 do
    g^(t) ← X^⊤ (X θ^(t) − y)
    θ^(t+1) ← θ^(t) − η g^(t)
    if ‖g^(t)‖ < τ then
        break
    end if
end for
return θ^(t)
```

As we can see from the tables, the coefficients and R^2 values are the same for OLS and GD. The coefficients for the interaction term are also the same. This is because the gradient descent algorithm converges to the same solution as OLS, which is the least squares solution. The R^2 values are also the same for both models, indicating that the models fit the data equally well.

(e) Perform arts (a), (b) and (c) above now using `stochastic gradient descent` (with one sample in each iteration). How does your result compare to the result from previous parts ? Note: while running `stochastic gradient descent`, you can sample without replacement and when you run out of samples, just start over. Note that you have to set the step-size parameter appropriately for this method.

	OLS	OLS, SGD	Interaction	Interaction, SGD
X1	-0.15147017	-0.15070764	-0.32877205	-0.32410012
X2	0.00773629	0.00874859	-0.23326364	-0.22732823
X3	0.79183280	0.79271560	0.77192750	0.77209111
X4	-0.04514359	-0.04433098	-0.04491517	-0.04496272
X1X2			0.38964848	0.38101380

Table 6: Estimated coefficients under various models (with SGD updates)

Model	Train	Test
OLS	0.51011	0.50499
OLS, SGD	0.50912	0.50294
Interaction	0.51735	0.51036
Interaction, SGD	0.51469	0.50820

Table 7: R^2 for SGD

We apply the stochastic gradient descent (SGD) algorithm to minimize the finite-sum

problem

$$\beta^* = \arg\min_{\beta \in \mathbb{R}^4} \frac{1}{n} \sum_{i=1}^{n} (y_i - x_i^\top \beta)^2 = \arg\min_{\beta \in \mathbb{R}^4} \frac{1}{n} \sum_{i=1}^{n} f_i(\beta).$$

At each iteration t, we pick one data point (x_i, y_i) (without replacement) and update

$$\beta^{(t+1)} = \beta^{(t)} - \eta_t \nabla f_i\big(\beta^{(t)}\big) = \beta^{(t)} - \eta_t \big(2\, x_i^\top x_i\, \beta^{(t)} - 2\, x_i^\top y_i\big).$$

As we can see from the tables, the coefficients and R^2 values are different for OLS and SGD. The coefficients for the interaction term are also different. This is because the stochastic gradient descent algorithm uses a different sample in each iteration, which can lead to different estimates of the coefficients. In our algorithm, the batch size is 1, which means that we are updating the coefficients after each sample. This can lead to more variability in the estimates of the coefficients compared to OLS, which uses all the samples at once. The R^2 value for OLS is higher than that for SGD, which suggests that OLS is a better fit for the data. However, the R^2 value for the interaction model with SGD is still quite high, indicating that it is still a good fit for the data.

3. (**15 Points**) Prove the Fact in Page 91 of `Opt.pdf` and solve the recursion in Page 92 to obtain the final result of the Theorem in Page 86. (**Hint:** You can use induction)

Solution: By μ-strong convexity, for any θ_1, θ_2 we have

$$f(\theta_2) - f(\theta_1) \;\geq\; \nabla f(\theta_1)^\top(\theta_2 - \theta_1) \;+\; \frac{\mu}{2}\|\theta_2 - \theta_1\|_2^2, \tag{3}$$

$$f(\theta_1) - f(\theta_2) \;\geq\; \nabla f(\theta_2)^\top(\theta_1 - \theta_2) \;+\; \frac{\mu}{2}\|\theta_2 - \theta_1\|_2^2. \tag{4}$$

Adding (3) and (4), we get

$$0 \;\geq\; \nabla f(\theta_1)^\top(\theta_2 - \theta_1) \;+\; \nabla f(\theta_2)^\top(\theta_1 - \theta_2) \;+\; \mu\,\|\theta_2 - \theta_1\|_2^2.$$

Observe that $\nabla f(\theta_2)^\top(\theta_1 - \theta_2) = -\,\nabla f(\theta_2)^\top(\theta_2 - \theta_1)$. Thus the above becomes

$$0 \;\geq\; \big[\nabla f(\theta_1) - \nabla f(\theta_2)\big]^\top(\theta_2 - \theta_1) \;+\; \mu\,\|\theta_2 - \theta_1\|_2^2.$$

Rearranging and changing sign,

$$\big[\nabla f(\theta_1) - \nabla f(\theta_2)\big]^\top(\theta_1 - \theta_2) \;\geq\; \mu\,\|\theta_2 - \theta_1\|_2^2,$$

which is the same as the inequality in the Fact. □

Now we start solving the recursion.

Assume $M_g = 0$ and let $\eta_t = \frac{c}{t+1}$, $c > 0$. Then, for the strong convexity $\nabla f(\theta^*) = 0$, we have the one-step bound

$$\mathbb{E}\big\|\theta^{(t+1)} - \theta^*\big\|_2^2 \;\leq\; \big(1 - 2\mu\eta_t\big)\,\mathbb{E}\big\|\theta^{(t)} - \theta^*\big\|_2^2 \;+\; \eta_t^2\,\sigma_g^2. \tag{5}$$

Base case: When $t = 0$, for (5), we have

$$\mathbb{E}\|\theta^{(1)} - \theta^*\|_2^2 \;\leq\; (1 - 2\mu c)\,\mathbb{E}\|\theta^{(0)} - \theta^*\|_2^2 \;+\; c^2\sigma_g^2. \tag{6}$$

Now pick $c = \mu^{-1}$, and define

$$c_0 \;:=\; \max\Big\{\,\mathbb{E}\|\theta^{(0)} - \theta^*\|_2^2,\; \frac{\sigma_g^2}{\mu^2}\Big\}.$$

Then (6) implies

$$\mathbb{E}\|\theta^{(1)} - \theta^*\|_2^2 \;\leq\; c_0.$$

Inductive step. Suppose for some $k \geq 1$, we have

$$\mathbb{E}\|\theta^{(k)} - \theta^*\|_2^2 \;\leq\; \frac{c_0}{k+1}. \tag{7}$$

We need to show

$$\mathbb{E}\|\theta^{(k+1)} - \theta^*\|_2^2 \;\leq\; \frac{c_0}{k+2}.$$

From (5) with $t = k$ and $\eta_k = c/(k+1)$, we have

$$\mathbb{E}\|\theta^{(k+1)} - \theta^*\|_2^2 \;\leq\; \Big(1 - 2\mu\frac{c}{k+1}\Big)\,\mathbb{E}\|\theta^{(k)} - \theta^*\|_2^2 \;+\; \frac{c^2\sigma_g^2}{(k+1)^2}.$$

Plugging in $c = \mu^{-1}$ and the inductive hypothesis (7), plus the fact $\sigma_g^2/\mu^2 \le c_0$, gives us

$$\begin{aligned}\mathbb{E}\|\theta^{(k+1)} - \theta^*\|_2^2 \;&\le\; \left(1 - \tfrac{2}{k+1}\right)\frac{c_0}{k+1} \;+\; \frac{c_0}{(k+1)^2} \\ &=\; \frac{c_0}{k+1} \;-\; \frac{2c_0}{(k+1)^2} \;+\; \frac{c_0}{(k+1)^2} \;=\; \frac{c_0}{k+1} \;-\; \frac{c_0}{(k+1)^2}.\end{aligned}$$

Since

$$\frac{c_0}{k+1} \;-\; \frac{c_0}{(k+1)^2} = \frac{k}{(k+1)^2}\,c_0 \;\le\; \frac{c_0}{k+2},$$

we have the desired bound

$$\mathbb{E}\|\theta^{(k+1)} - \theta^*\|_2^2 \;\le\; \frac{c_0}{k+2}.$$

By induction on $t = 0, 1, 2, \ldots$, it follows that for all $t \ge 0$,

$$\boxed{\mathbb{E}\left\|\theta^{(t)} - \theta^*\right\|_2^2 \;\le\; \frac{c_0}{t+1}.}$$

□

4. (**10 Points**) In class, we defined the notion of a sub-gradient and worked out the example of the absolute function ($f(\theta) = |\theta|$).

   Consider the function from $\mathbb{R}^d \to \mathbb{R}$ which is the Euclidean norm for a vector, i.e., $f(\theta) := \|\theta\|_2$. Compute the sub-gradient of this function.

   Solution: We know that $g \in \partial f(\theta)$ if and only if

$$f(y) \geq f(\theta) + g^\top (y - \theta) \quad \forall y \in \mathbb{R}^d.$$

   Case 1: $\theta \neq 0$**.** Since f is differentiable at any $\theta \neq 0$, we have

$$\nabla f(\theta) = \nabla \|\theta\|_2 = \frac{\theta}{\|\theta\|_2}.$$

   Case 2: $\theta = 0$**.** By definition, $g \in \partial f(0)$ if and only if

$$f(y) \;\geq\; g^\top (y - 0) \quad \Longleftrightarrow \quad \|y\|_2 \;\geq\; \langle y,\, g\rangle \quad \forall y \in \mathbb{R}^d.$$

   By the Cauchy–Schwarz inequality, we have

$$\langle y, g\rangle \leq \|y\|_2 \, \|g\|_2,$$

   so the above holds for every y precisely when

$$\|g\|_2 \leq 1.$$

   Conversely, if $\|g\|_2 > 1$ then choosing y parallel to g with a sufficiently large norm violates the inequality. Combining these two cases,

$$\boxed{\partial \|\theta\|_2 = \begin{cases} \dfrac{\theta}{\|\theta\|_2}, & \theta \neq 0, \\ \{\, g \in \mathbb{R}^d : \|g\|_2 \leq 1\}, & \theta = 0. \end{cases}}$$

Pledge:
Please sign below (print full name) after checking (✓) the following. If you can not honestly check each of these responses, please email me at kbala@ucdavis.edu to explain your situation.

- We pledge that we are honest students with academic integrity and we have not cheated on this homework. (✓)
- These answers are our own work. (✓)
- We did not give any other students assistance on this homework. (✓)
- We understand that to submit work that is not our own and pretend that it is our is a violation of the UC Davis code of conduct and will be reported to Student Judicial Affairs.
- We understand that suspected misconduct on this homework will be reported to the Office of Student Support and Judicial Affairs and, if established, will result in disciplinary sanctions up through Dismissal from the University and a grade penalty up to a grade of "F" for the course. (✓)

Name: Adrian Zhou